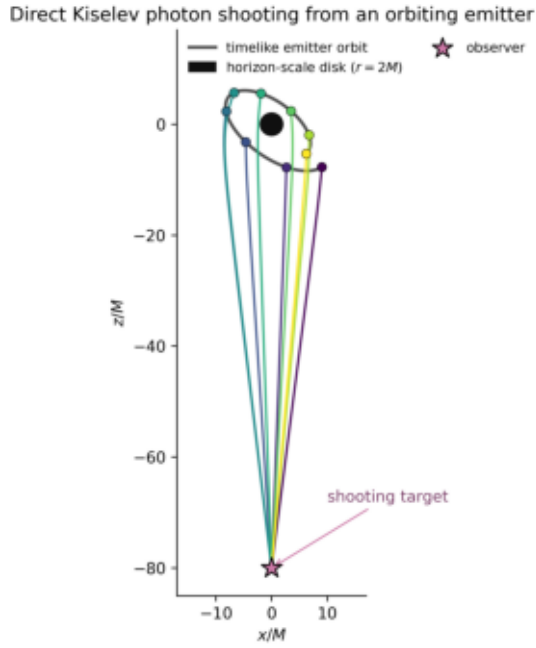
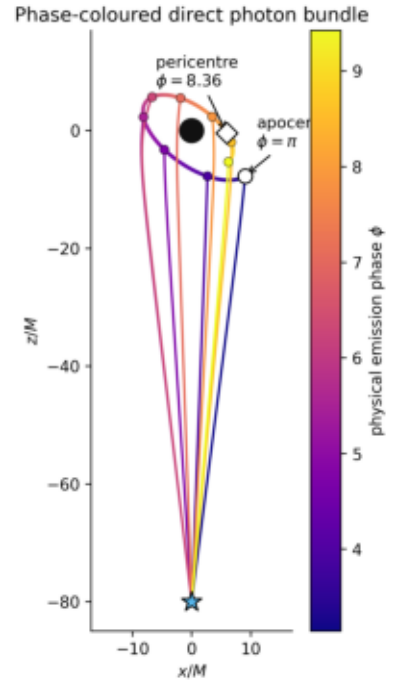


physical shooting geometry overview

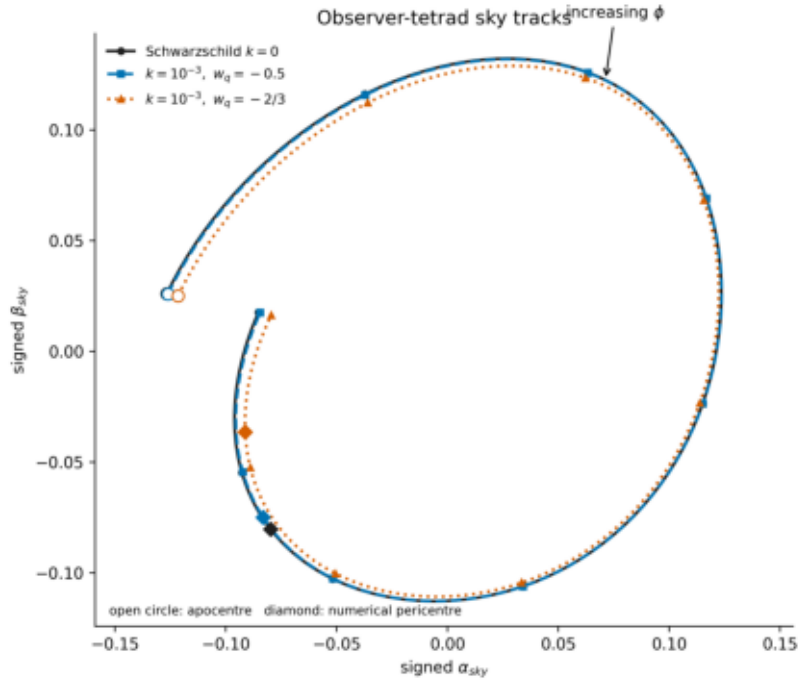


Initial photon directions are adjusted until every direct-branch ray reaches the observer.

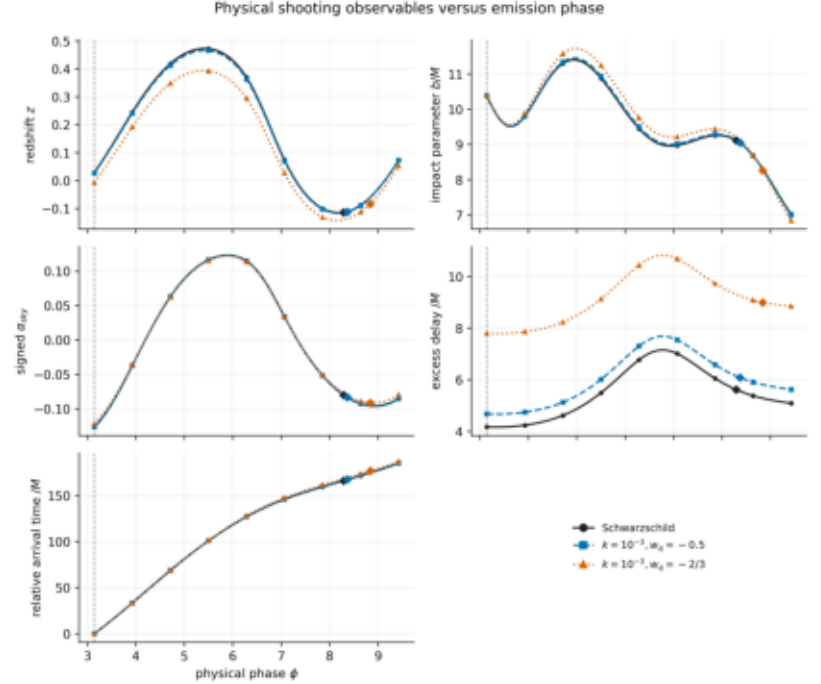
phase coloured photon shooting



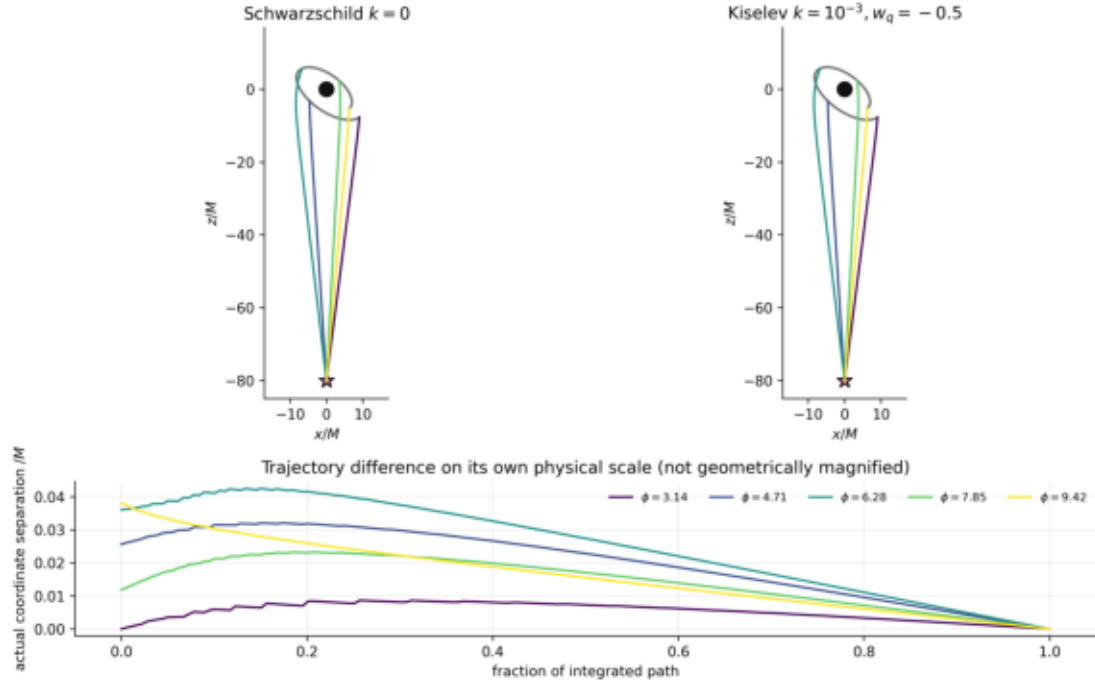
observer sky track comparison



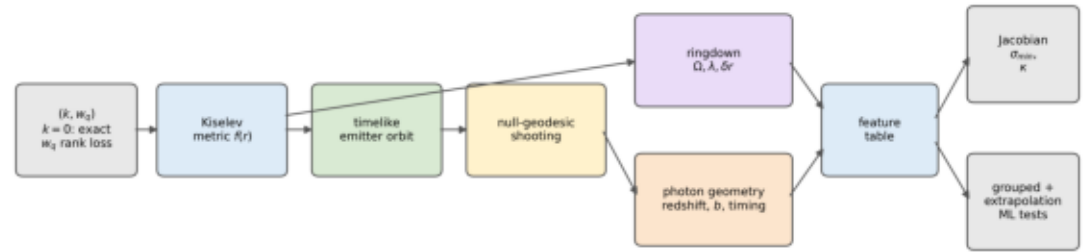
shooting observables vs phase



schwarzschild kiselev ray comparison



physical shooting to ml pipeline



Complementary observable families are combined only after physical validation.

physical shooting poster composite

Physical geodesic shooting links spacetime parameters to complementary observables

